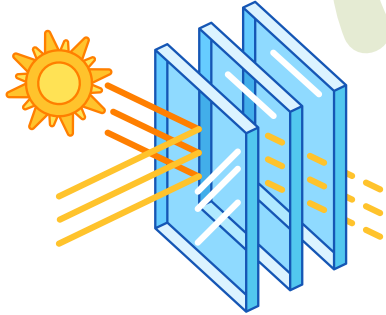


Green Marine Glass Solutions

High-Performance Energy-Efficient Glass for Marine Applications



- Low-E (Low Emissivity) Coatings
- Insulating Glass
- Energy efficiency + Cost reduction
- U Value $W/(m^2K) \leq 1.1$

**Up to 30% less HVAC
energy with KARACA
glazing systems**



What Is U-Value and Why Does It Matter?

- U-value measures how well a glass unit insulates.
- Lower U-value = Better thermal insulation = Less energy loss
- Helps maintain interior comfort in both cold and hot climates.
- Reduces HVAC energy consumption on board vessels.



Best Applied In:

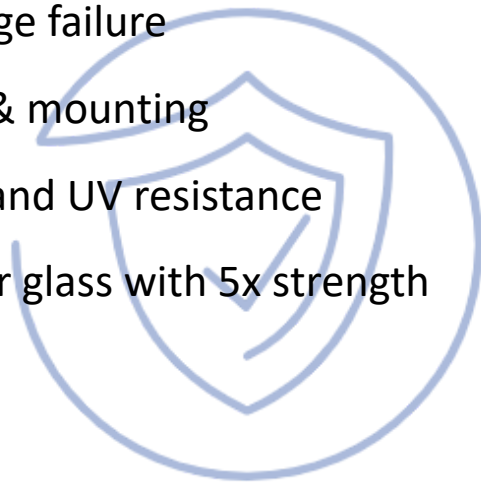
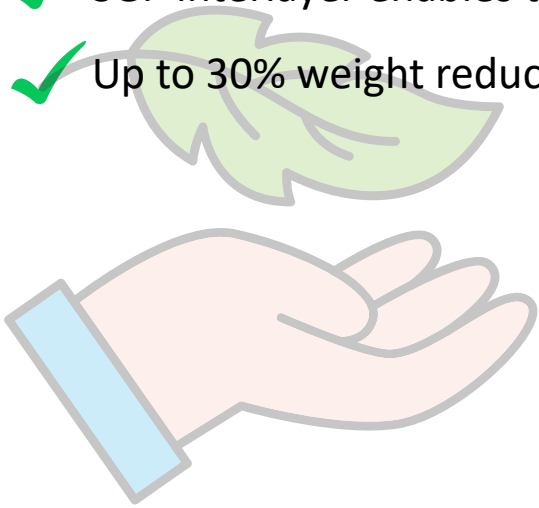
- Bridge / Wheelhouse
- Crew accommodation
- VIP passenger cabins
- Areas with HVAC or tight climate control requirements

Lightweight & High-Strength Glass Solutions

SGP Interlayer Advantage + Thin Glass Configurations

Technical Advantages

- ✓ Strength: Resists impact & edge failure
- ✓ Lightweight: Easier handling & mounting
- ✓ Durability: Superior weather and UV resistance
- ✓ SGP interlayer enables thinner glass with 5x strength
- ✓ Up to 30% weight reduction



KARACA
SHIP WINDOWS & WIPERS